

Section 1.1: Systems of Linear Equations - Worksheet

MATH 3260 – Linear Algebra

Part A: Concepts and classification

- Decide whether each equation is linear. Explain any nonlinear classification.
 - $3x - 4y + z = 21$
 - $xy + z = 9$
 - $x_1 - 2x_2 = 0$
 - $\frac{1}{x} + y = 4$
 - $5u - 2v + 7w = 0$
 - $\sin x + y = 3$
- Explain in one sentence what it means for (a, b) to solve a system of two equations.
- Check whether $(4, 5)$ solves both $x + y = 9$ and $2x - y = 3$.

Part B: Classifying solution sets

- Solve and classify:

$$\begin{cases} x + y = 20, \\ x - y = 2. \end{cases}$$

- Classify without graphing:

$$\begin{cases} 2x + 4y = 20, \\ x + 2y = 12. \end{cases}$$

- Write the complete solution set in parametric form:

$$x - y + 2z = 9.$$

- Sketch or describe the geometry of each system from Problems 4–6.

Part C: Elimination and applications

1. Solve by elimination. Show each intermediate system or augmented matrix.

$$\begin{cases} x + y + z = 30, \\ 2x + y - z = 19, \\ x - y + 2z = 12. \end{cases}$$

2. Two quantities u and v have total 21, and u is 3 less than v . Write and solve a system.
3. A two-category plan contains 21 items total, and the weighted count $x + 2y$ equals 30. Find x and y .
4. A fictional arena bundle contains two drinks and one snack for 21 dollars, while one drink and two snacks cost 24 dollars. Find the modeled item prices.
5. Challenge: construct your own two-equation system having infinitely many solutions and explain why.

Section 1.2: Gaussian and Gauss-Jordan Elimination - Worksheet

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Part A: Matrix language

1. State the dimensions and identify a_{23} for

$$A = \begin{bmatrix} 1 & 9 & 20 \\ 21 & 4 & 7 \end{bmatrix}.$$

2. Write the coefficient matrix and augmented matrix for

$$\begin{cases} x - 2y + z = 9, \\ 3x + y - 4z = 20. \end{cases}$$

3. Apply $R_2 \leftarrow R_2 - 3R_1$ to

$$\left[\begin{array}{cc|c} 1 & 2 & 9 \\ 3 & 7 & 30 \end{array} \right].$$

4. State whether each matrix is REF, RREF, or neither:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

Part B: Gaussian and Gauss-Jordan elimination

1. Solve by Gaussian elimination, showing every row operation:

$$\begin{cases} x + y + z = 21, \\ 2x + y + 3z = 41, \\ x + 3y + 2z = 38. \end{cases}$$

2. Reduce to RREF:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 20 \\ 1 & 2 & 1 & 42 \\ 2 & 1 & 1 & 40 \end{array} \right].$$

Part C: Structure and applications

1. Classify and solve:

$$\begin{cases} x + 2y - z = 9, \\ 2x + 4y - 2z = 18. \end{cases}$$

2. Find all solutions of the homogeneous system

$$x + y + z = 0, \quad 2x + 2y + 2z = 0.$$

3. A fictional Spurs fan shop sells a cap c , a shirt s , and a poster p :

$$\begin{cases} r + s + g = 50, \\ 2r + s + 3g = 111, \\ r + 2s + 4g = 119. \end{cases}$$

Solve using Gaussian elimination.

4. Explain the meaning of each row:

$$[0 \ 0 \ 0 \mid 0], \quad [0 \ 0 \ 0 \mid 5].$$

5. Challenge: give an example of a homogeneous system in three variables with only the trivial solution.

Section 1.3: Applications of Linear Systems - Worksheet

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Part A: Curve fitting

1. A quantity-remaining model records $(1, 130)$ and $(4, 40)$. Find the line $y = mx + b$ and interpret the slope.
2. Find the quadratic $p(x) = ax^2 + bx + c$ passing through
$$(0, 1), \quad (1, 9), \quad (2, 21).$$

Show the coefficient system and row-reduction steps.

3. A check-in count is modeled by a quadratic through $(0, 20)$, $(1, 29)$, and $(2, 36)$. Find the quadratic.

Part B: Network flow

1. A node receives 50 vehicles per hour and sends flows x_1 and x_2 outward. Write the conservation equation.
2. Analyze the model

$$x_1 + x_2 = 50, \quad x_1 + 9 = x_3, \quad x_2 + x_3 = 59.$$

Write the flow family using $x_1 = t$ and determine the feasible range for nonnegative flows.

3. The same model is reported with the final total changed from 59 to 70. Explain what happens and why model checking matters.

Part C: Circuits, calibration, and planning

1. Solve the electrical-circuit model, showing elimination:

$$I_1 - I_2 + I_3 = 0, \quad I_1 + I_2 = 21, \quad I_2 + 2I_3 = 20.$$

2. A color-sensor calibration model satisfies

$$r + g + b = 21, \quad 2r + g + b = 30, \quad r + 2g + 3b = 44.$$

Find (r, g, b) and verify the result.

3. A resource-allocation model uses u, v, w :

$$u + v + w = 9, \quad 2u + v = 12, \quad v + 2w = 8.$$

Solve and interpret.

4. State one assumption needed for each of the following to be linear: a traffic network, a circuit model, and a curve-fitting model.